

Data Management Standards & Guidelines

Organization: Company A (Government Entity)
Document ID: DM-STD-001
Version: 2.5
Classification: Internal
Date: March 1, 2024
Total Pages: 26

DOCUMENT CONTROL

Document Information

Attribute	Details
Document ID	DM-STD-001
Document Title	Data Management Standards & Guidelines
Version	2.5
Status	Active / Enforced
Classification	Internal
Owner	Data Management Office (DMO)
Author	DMO Technical Team
Effective Date	March 1, 2024
Review Cycle	Annual

Version History

Version	Date	Author	Description of Changes	Verified By
1.0	2023-01-20	Data Architect	Initial Standards for Database Naming.	QA Lead
2.0	2023-06-15	DQ Lead	Added Data Quality dimensions and metrics.	DMO Manager
2.5	2024-03-01	Architect	Comprehensive update covering modeling and metadata.	QA Lead

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1. INTRODUCTION

1.1 Purpose

The Data Management Standards provide the "technical teeth" to the Data Governance Policy. While the Policy says "what" to do (e.g., "Data must be high quality"), these Standards say "how" to do it (e.g., "Dates must be stored as ISO 8601").

Adherence to these standards ensures:

- Interoperability: Systems can talk to each other because they speak the same language.
- Maintainability: Developers can understand each other's code and databases.
- Quality: Errors are prevented by enforcing structure.

1.2 Audience

This document is mandatory for:

- Data Architects: Designing new systems.
- Database Administrators (DBAs): Creating tables and columns.
- Developers: Writing code that interacts with data.
- Data Stewards: Defining business terms.

2. DATA NAMING CONVENTIONS

2.1 General Principles

1. English Language: All technical names (Tables, Columns) must be in English.
2. Meaningful: Names must describe the content. Avoid generic names like DATA_TABLE_01.
3. No Spaces: Use underscores _ or CamelCase depending on the object type. Spaces are strictly forbidden.
4. No Reserved Words: Do not use SQL reserved words (e.g., SELECT, FROM, DATE, USER).

2.2 Relational Database Objects

2.2.1 Tables

- Syntax: `[Prefix]\_[SubjectSubject]\_[Type]`
- Case: PascalCase or UPPER_SCREAMING_SNAKE_CASE (depending on DB platform). Oracle Standard: UPPER_CASE with Underscores.
- Prefixes:
 - `TBL\_`: Standard Table
 - `REF\_`: Reference/Lookup Table
 - `STG\_`: Staging Table
 - `DIM\_`: Dimension Table (Warehouse)
 - `FACT\_`: Fact Table (Warehouse)
- Examples:
 - `TBL\_EMPLOYEE` (Correct)
 - `REF\_COUNTRY\_CODES` (Correct)
 - `EmpInfo` (Incorrect - unclear prefix/case)

2.2.2 Views

- Syntax: `VW\_[BusinessName]`
- Example: `VW\_ACTIVE\_EMPLOYEES`

2.2.3 Keys

- Primary Key: `PK\_[TableName]`
- Foreign Key: `FK\_[SourceTable]\_[TargetTable]`
- Index: `IX\_[TableName]\_[ColumnName]`

2.3 File Naming Standards

For unstructured files (Excel, PDF, CSV) stored in shared repositories:

- Syntax: `[Project/Dept]\_[ContentType]\_[Date]\_[Version].[Ext]`
- Date Format: `YYYYMMDD` (Critical for sorting).
- Version: `vX.Y`
- Examples:
 - `HR\_PayrollReport\_20240301\_v1.0.xlsx` (Correct)
 - `meeting notes.docx` (Incorrect - no context, space, no date)
 - `ProjectAlpha\_Final\_Final\_V2.pdf` (Incorrect - ambiguous versioning)

2.4 Attribute/Column Naming

- Syntax: `[Entity]\_[Attribute]`
- Abbreviation: Use standard abbreviations from Appendix A (e.g., `DTE` for Date, `AMT` for Amount).

- Suffixes:
- `\_ID`: Identifier (Primary Key)
- `\_CODE`: Business Code (e.g., `SA`)
- `\_DESC`: Description
- `\_DTE`: Date
- `\_FLAG`: Boolean (Y/N or 0/1)
- `\_AMT`: Monetary Amount
- `\_QTY`: Quantity/Count
- Examples:
- `EMP\_FIRST\_NAME`
- `EMP\_HIRE\_DTE`
- `ORDER\_TOTAL\_AMT`
- `IS\_ACTIVE\_FLAG`

3. DATA MODELING STANDARDS

3.1 Conceptual Data Models

- Definition: High-level view for business stakeholders.
- Requirement: Must be created for every Project Initiation Document (PID).
- Content:
- Key Entities (e.g., "Patient", "Appointment").
- Relationships (e.g., "Patient" makes "Appointment").
- No attributes or keys shown.
- Tool: Visio or Sparx EA.

3.2 Logical Data Models

- Definition: Detailed view for application design.
- Requirement: Must be approved by Data Architect before Build phase.
- Content:
- All Entities and Attributes.
- Primary and Foreign Keys defined.
- Normalization: 3rd Normal Form (3NF) required for transactional systems.
- Business definitions for every attribute.

3.3 Physical Data Models

- Definition: DB-specific schema generation.

- Requirement: Auto-generated from Logical Model.
- Content:
- Data Types (e.g., `VARCHAR2(50)`).
- Nullability constraints.
- Partitioning strategy.
- Storage parameters.

3.4 Modeling Tools

- Mandatory Tool: Company A uses Sparx Enterprise Architect as the central repository.
- Versioning: Models must be version controlled. Changes must be tracked.

4. METADATA STANDARDS

4.1 Business Metadata

This data describes the "meaning" and is maintained in the Informatica Data Catalog.

Mandatory Fields for All Assets:

1. Name: Unique Business Name.
2. Description: Detailed definition (Min 20 chars).
 - Bad: "The employee date."
 - Good: "The date on which the employee officially joined the organization as per the contract."
3. Data Owner: Name of Accountable Exec.
4. Data Steward: Name of Responsible SME.
5. Classification: Public / Internal / Confidential / Secret.
6. Security Tag: PII / PHI / PCI (if applicable).

4.2 Technical Metadata

Captured automatically by scanners.

1. Physical Name: TBL_HR_01
2. Path: Server1/SchemaHR
3. Profile Stats: Row count, Null count, Distinct values.

4.3 Operational Metadata

1. Last Refreshed: Timestamp of last ETL load.
2. Lineage: Upstream source and downstream consumption.

5. DATA QUALITY STANDARDS

5.1 Standard Dimensions

Company A measures quality using the 6 standard dimensions:

Dimension	Definition	Example Rule
Completeness	Data is present.	National_ID cannot be NULL.
Accuracy	Data reflects reality.	DOB cannot be in the future.
Consistency	Data matches across systems.	Gender in HR must match Gender in Payroll.
Timeliness	Data is available when needed.	Dashboard data < 24 hours old.
Validity	Data follows format/domain.	Email must contain '@'.
Uniqueness	No duplicates.	Emp_ID must be unique.

5.2 Mandatory DQ Rules for CDEs

For Critical Data Elements (CDEs), specific rules are mandatory:

- Dates: All dates must be validated for format and logical ranges (e.g., No birth dates before 1900).
- Emails: Must pass regex validation `^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`.
- Phone Numbers: Must follow E.164 format (+9665xxxxxxxx).
- National IDs: Must pass the checksum algorithm (Luhn algorithm or equivalent Saudi ID check).

5.3 Thresholds & Tolerances

- Critical Systems:
 - Red (Fail): < 95%
 - Amber (Warning): 95% - 98%
 - Green (Pass): > 98%
- Non-Critical Systems:
 - Red (Fail): < 80%
 - Amber (Warning): 80% - 90%
 - Green (Pass): > 90%

6. REFERENCE & MASTER DATA STANDARDS

6.1 Standard Reference Codes

To ensure consistency, we use international standards:

- Country Codes: ISO 3166-1 Alpha-2 (e.g., `SA`, `US`, `AE`).
- Currency Codes: ISO 4217 (e.g., `SAR`, `USD`).
- Language Codes: ISO 639-1 (e.g., `ar`, `en`).
- Date Format: ISO 8601 (`YYYY-MM-DD`).

Internal Standard Reference Lists:

- Located in the Reference Data Management (RDM) tool.
- Example: `REF\_ORG\_UNITS` (List of all valid departments).

6.2 Master Data Golden Records

- Single Source of Truth: Data for core entities must come from the designated master.
- Customer/Citizen: Mastered in CRM.
- Employee: Mastered in SAP HR.
- Vendor: Mastered in Oracle ERP.
- Golden Record ID: A unique Universal ID (UUID) must be generated for every master record and persisted across systems.

END OF DOCUMENT

Company A - Internal Use Only